General Mathematics — Exercise Sheet 33

First-order differential equations

Variant: Statements

Exercise 1 (Logistic growth and its interval of solution). Let $\rho, K > 0$ and $y_0 \in \mathbb{R}$. Solve $y' = \rho y(1 - y/K)$, $y(0) = y_0$. Find the largest open interval containing zero on which the solution is finite. Distinguish $y_0 < 0$, $y_0 = 0$, $0 < y_0 < K$, $y_0 = K$ and $y_0 > K$. For nonnegative initial values, find the future limit. Use Lecture 33's separation and linear uniqueness, and Lecture 8's logarithm rules. Prove uniqueness directly by writing a linear equation for the difference of two solutions; do not invoke the cited existence theorem.

Exercise 2 (Two models of downward drag). Take downward as positive. A mass $m > 0$ falls from rest under gravity $g > 0$. Its speed v satisfies one of the prescribed models

$$\text{linear drag: } mv' = mg - kv \ (k > 0), \qquad \text{quadratic drag: } mv' = mg - cv^2 \ (c > 0).$$

For each model find $v(t)$ for $t \geq 0$, prove it stays nonnegative, and find its limiting speed. For quadratic drag also find downward distance $s(t)$ with $s(0) = 0$ and $s' = v$. Use Lecture 33's linear or separation method and Lecture 23's primitives. Prove uniqueness for the quadratic solution through $v(0) = 0$ by a linear equation for the difference. The quadratic model here is prescribed only for downward motion.

Exercise 3 (Mixing in a tank whose volume changes). A tank initially contains 100 litres of well-mixed solution and 50 grams of salt. Solution containing 2 grams per litre enters at 3 litres per minute. The well-mixed contents leave at 1 litre per minute. Assume unlimited tank capacity, and let $Q(t)$ be the salt mass in grams for $t \geq 0$ minutes. Form the mass-balance equation, solve it, and find the salt mass after 50 minutes. Prove the concentration stays between zero and the incoming concentration and find its future limit. Use Lecture 33's linear formula and Lecture 8's derivative rules.

Exercise 4 (Bernoulli's substitution on positive solutions). Let $p, q : I \to \mathbb{R}$ be continuous on an open interval and let $\alpha \in \mathbb{R} \setminus \{0, 1\}$. Show that positive solutions of $y' + p(t)y = q(t)y^\alpha$ correspond to positive solutions of a linear equation under $u = y^{1-\alpha}$. State that equation and its converse transformation. Apply the method to

$$y' + y = e^t y^2, \qquad y(0) = \frac{1}{2},$$

and give its largest positive-solution interval containing zero. Check the zero solution separately for this particular equation. Explain how the excluded cases $\alpha = 0$ and $\alpha = 1$ are already linear. Use Lecture 8's real-power chain rule and Lecture 33's linear formula.

Exercise 5 (A resistor–capacitor circuit with sinusoidal voltage). Assume the circuit model

$$RQ'(t) + \frac{Q(t)}{C} = V_0 \cos(\omega t), \qquad Q(0) = Q_0,$$

where $R, C > 0$, $\omega \geq 0$, and $V_0, Q_0 \in \mathbb{R}$. The unknown Q is capacitor charge and the current is $I = Q'$. Use Lecture 33's integrating-factor formula to find Q and I. Identify the term which decays as $t \to \infty$, and treat $\omega = 0$ explicitly. Use Lecture 23's primitives and Lecture 8's trigonometric derivatives. No complex circuit notation is assumed.